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Data Science Fundamentals

Lasso Regression

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5		Week - 13	Regularization: Lasso regression
Week - 6		Week - 14	Recap: Theory + Excercises
Week - 7		Week - 15	Exam Preperation + Q&A
Week -			



(subject to change)

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Now we get to the fun part

LASSO Regression

Regression again, really?

- Well yeah, but it is more than that!
- We can use one ML algortihm for another purpose
- Where have we seen this before?
- Logistic regression!
- Just like then, we will be using a **regression** model for another purpose: **feature selection**!



Feature selection

- Selecting the **most important** (or least redundant) features of a dataset
- Reduces **size** and **variety** of dataset
- Improves processing time when creating ML models
- Finding a balance between dataset size and model accuracy
- **Similar** goals to PCA but remember PCA =/= feature selection!
- PCA **creates** new features, feature selection only **drops** redundant ones.



Lasso regression

- **L**east **A**bsolute **S**hrinkage and **S**election **O**perator
- Takes a base linear or polynomial regression model
- Adjusts the coefficients using **penalization**



Penalization

- Adding a '**penalty**' term to punish complex models
- Promotes simple models that catch a **broader sense** of the dataset
- The **higher** your penalty, the **simpler** your final model
- → be careful for underfitting!
- Especially useful in models with **a lot** of features/dimensions, as they can easily get complex

Inner workings of lasso regression

- Start with linear regression: minimize prediction errors
- Add a penalty: shrink **large coefficients** down
- Penalty strength controlled by parameter '**alpha**' (α)
- Lasso forces **irrelevant features'** coefficients to **zero**
- Important features stay; unimportant ones are removed
- Final model is **simpler** and **avoids overfitting**.



Example

- Let's start with the base linear regression formula

$$f(x) = ax + b$$

Finding the best alpha

- Now that we know how penalization works, how do we pick our penalty term?
- Twe methods:
- **Visually** identify it by drawing a plot
  - Loop over values of alpha, pick the best one
  - More **intuitive** but hyper **specific**
- Using **Grid Search Cross validation**
  - Just call a method and let python handle it :P
  - We only need to pass on a **scoring parameter**
  - Less intuitive but **better general result**



Actual feature selection

- All of this gives us a **standalone lasso regression model**
- Great, but not exactly what we want
- We want to use it for **feature selection**
- How do we use the created model to decide which features to drop from our dataset?

We look at the coefficients of our model!

Features where the coefficient is **zero** are **dropped**!

Lasso regression outside of regression

- Lasso regression is inherently a **regression** model, but it can still be used elsewhere
- How? By making use of its **feature selection** power!
- This only applies when trying to predict something specific ➔ **supervised learning**



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Let's jump into action

Exercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 10 – Lasso Regression Notebook"
- Also download the csv file called "Advertising.csv" from Linear Regression
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 10 – Lasso Regression Exercise"
- Download the csv file called "Diabetes.csv" from Logistic Regression
- Also download the csv file called "Possum.csv"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises



I don't bite, ask me any questions you have!